

Digital Triage Pseudocode

Emergency Department Patient Risk Stratification System

CariSurg MedTech Pathways | Week 0 Final Deliverable

Executive Summary

This document presents a pseudocode design for a digital triaging system that uses blood pressure and oxygen saturation (SpO₂) to categorize patients into risk levels. The system follows the Emergency Severity Index (ESI) framework and provides clear decision logic for clinical implementation.

1. System Overview

The digital triage system processes patient vital signs (SBP, DBP, SpO₂) and assigns a triage category from Level 1 (Critical) to Level 5 (Non-Urgent). The algorithm prioritizes critical conditions first, ensuring immediate identification of life-threatening situations.

2. Input Parameters

Parameter	Unit	Normal Range	Description
Systolic BP	mmHg	90-120	Peak arterial pressure
Diastolic BP	mmHg	60-80	Minimum arterial pressure
SpO ₂	%	95-100	Oxygen saturation

Table 1: Input vital sign parameters

3. Clinical Risk Thresholds

3.1 Blood Pressure Classification

Category	SBP (mmHg)	DBP (mmHg)	Risk
Severe Hypotension	< 80	< 50	Critical
Moderate Hypotension	80-89	50-59	Emergent
Normal	90-120	60-80	Low
Hypertensive Crisis	>= 180	>= 120	Critical

Table 2: Blood pressure thresholds

3.2 Oxygen Saturation Classification

Category	SpO ₂ (%)	Clinical Significance	Risk
Severe Hypoxemia	< 85	Critical hypoxia; organ damage risk	Critical
Moderate Hypoxemia	85-89	Significant hypoxia; intervention needed	Emergent
Mild Hypoxemia	90-94	Below normal; monitoring required	Urgent

Normal	95-100	Adequate oxygenation	Low
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Table 3: Oxygen saturation thresholds

4. Triage Algorithm Pseudocode

4.1 Main Triage Function

```
FUNCTION calculate_triage_level(sbp, dbp, spo2): // Input validation IF any input is NULL: RETURN "ERROR"
bp_risk = assess_blood_pressure(sbp, dbp)
oxygen_risk = assess_oxygen_saturation(spo2)
overall_risk = MAX(bp_risk, oxygen_risk)
RETURN map_to_triage_category(overall_risk)
```

4.2 Blood Pressure Assessment

```
FUNCTION assess_blood_pressure(sbp, dbp): IF sbp < 80 OR dbp < 50: RETURN CRITICAL
IF sbp >= 180 OR dbp >= 120: RETURN CRITICAL
IF sbp 80-89 OR dbp 50-59: RETURN EMERGENT
IF sbp >= 130 OR dbp >= 80: RETURN MODERATE
RETURN LOW
```

4.3 Oxygen Saturation Assessment

```
FUNCTION assess_oxygen_saturation(spo2): IF spo2 < 85: RETURN CRITICAL
IF spo2 85-89: RETURN EMERGENT
IF spo2 90-94: RETURN URGENT
RETURN LOW
```

5. Triage Category Output

Level	Category	Max Wait	Color
1	Resuscitation	0 min	RED
2	Emergent	10 min	ORANGE
3	Urgent	30 min	YELLOW
4	Less Urgent	60 min	GREEN
5	Non-Urgent	120 min	BLUE

Table 4: ESI triage categories

6. Conclusion

This pseudocode provides a systematic approach to digital triage using blood pressure and oxygen saturation. The algorithm uses established clinical thresholds and the ESI framework to categorize patients by urgency level.